06

FDP DAY 6 · 07 NOV 2026

Agents at work

AI agents and their applications in teaching and research

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SIX EVENINGS

Where we have been, and where tonight goes

DAY 1

AI tools

The generative AI landscape

What each tool is good for

✓ DONE

DAY 2

Prompting

Prompts for teaching, research and productivity

✓ DONE

DAY 3

Research and data

AI for research, data analysis and visualisation

✓ DONE

DAY 4

Build with AI

The harness

A form, rebuilt with a skill

NotebookLM

✓ DONE

DAY 5

Agents that plan

Spec Kit and Superpowers

An LLM wiki

Class prep

✓ DONE

DAY 6

Agents at work

Build the Class Prep Studio live

More applications

Using AI well

TONIGHT

Tonight: everything from the first five evenings, working together on one real task.

TONIGHT WE BUILD

The Class Prep Studio

IN

- a topic
- your textbook chapter
- your syllabus unit

OUT

- a lesson plan
- class slides
- important questions, with answers
- a practice quiz
- a student handout with diagrams

ASKED EVERY EVENING

Make it sound like you

GENERAL WRITING **humanizer**	`blader/humanizer`	Removes tell-tale AI phrasing and keeps your meaning. About 65,000 stars.
PAPERS AND PROPOSALS **academic-humanizer**	`AIScientists-Dev/academic-humanizer`	Keeps scholarly precision; never changes your citations or data.
BEFORE YOU SUBMIT **check-citations**	`PHY041/claude-skill-citation-checker`	Checks every reference against CrossRef, Semantic Scholar and OpenAlex.

Use them as an editor, not the author.

Detectors are unreliable: they flag honest writers too. Don’t chase a score.

Disclose AI use the way your journal or institution asks.

GRILL ME

Why agents build the wrong thing

What you meant

"Slides for Unit 4, for my first-years"

the gap

What the agent built

Twenty generic slides, the wrong level, none of your examples

The fix: let the agent interview you before it starts.

GRILL ME

Matt Pocock's skills

github.com/mattpocock/skills

/grill-me	a relentless interview about your plan, until nothing is left unclear
/research	investigates a question against trusted sources; writes cited notes
/teach	teaches you a new concept over several sessions
/handoff	packs a long conversation into a note another agent can pick up

GRILL ME

How a grilling works

01 **Rounds.** It asks every question it can ask right now, numbered.

02 **A recommendation.** Each question comes with its suggested answer: say “yes”, or change it.

03 **Facts vs choices.** Facts it looks up itself; decisions it leaves to you.

04 **Done.** When nothing is left assumed, it writes the brief.

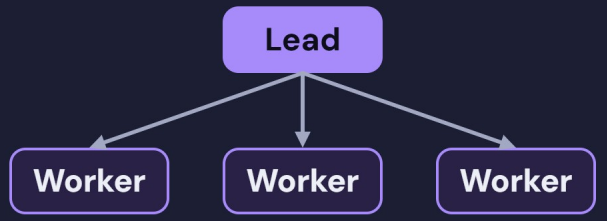
DEMO

The agent grills me

HOW AGENTS WORK TOGETHER

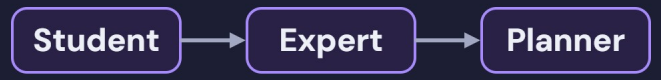
Five ways agents work together

Orchestrator + workers



In the studio: **runs the whole studio**

Pipeline



In the studio: **the debate**

Fan-out, fan-in



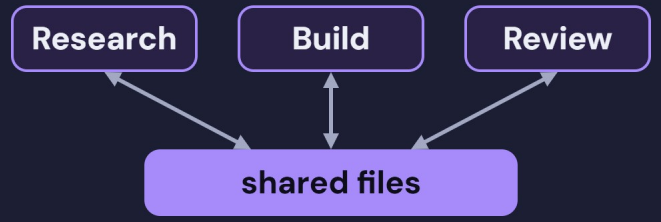
In the studio: **research and build**

Builder + critic loop



In the studio: **builders and reviewer**

Shared memory



In the studio: **one folder for everyone**

Tonight's studio uses all five

- ✓ Lead · the orchestrator
- ✓ Pipeline · the debate
- ✓ Fan-out · research, build
- ✓ Critic loop · the reviewer
- ✓ Shared files · one folder

How agents talk

shared files	brief.md, research/, plan.md, review.md
the orchestrator	“textbook agent, start now”
agent ↔ tools	MCP: one plug, many apps
agent ↔ agent	A2A (Agent2Agent): agents from different companies, the emerging standard

HOW AGENTS WORK TOGETHER

The telephone game

You	“threshold is 60%”
Agent 1	“threshold 60%”
Agent 2	“threshold 6%”
Agent 3	“Excellent news: every CO attained!”

FOUR FAILURE MODES

01 An early mistake flows into every later step

02 A reviewer agent agrees with the builder

03 Context is lost at a handover

04 Free-plan limits run out faster with several agents

THE PRD · PRODUCT REQUIREMENTS DOCUMENT

The brief an agent can build from

PRD.md

Problem
what hurts today

Goal
what “done” looks like

Users
who it is for

Inputs
what you give it

Outputs
the files you get

Workflow
the steps, in order

Agents
who does each step, and what each must never do

Gates
where you approve

Guardrails
data, copyright, nothing sent

THE PRD · SECTION 6

The studio on one page



Every arrow is a file you can open: brief.md · research/ · debate.md · plan.md · outputs/ · review.md

The agents, their tools, and what each must never do

Agent	Tools	Must never
Grill	none	ask more than 6 questions a round
Researchers ×3	read files; web search	use a source without a page number or link
Student + expert	read the research	answer from outside the research
Planner	read and write files	hide a conflict between sources
Builders ×4	pptx, pdf, education skills	add anything not in the approved plan
Reviewer	read only	edit any output
Orchestrator	start agents	skip a gate

DEMO

Hand the PRD to the agent

MORE APPLICATIONS

Six applications for your work

TEACHING

Question paper

`setter → reviewer → key`

✓ you approve the paper

RESEARCH

Literature review

`wiki → matrix → citations`

✓ you open the flagged citations

DEPARTMENT

CO attainment report

`marks → attainment → summary`

✓ you check one CO, then sign

TEACHING

Socratic tutor

`asks, never tells`

✓ grounded in your notes

RESEARCH

Reading digest

`new papers → your wiki, Mondays`

✓ reads and drafts only

DEPARTMENT

Meeting minutes

`transcript → minutes, actions`

✓ you correct, then send

MORE APPLICATIONS

Every application is tools plus permissions

Application	Tools it needs	Permission	The guard
Question paper	read syllabus; write files	its own folder only	works on a copy
Literature review	read PDFs; look up DOIs	one website, read only	only DOIs leave the laptop
CO attainment	read marks; calculate	a sandbox folder	synthetic marks only
Socratic tutor	read your notes	no web, nothing out	grounded in your notes
Reading digest	web search; write to wiki	scheduled; read, draft	never sends anything
Meeting minutes	read a transcript	drafts only	you send it, not the agent

MORE APPLICATIONS

Workflows on a schedule

READ AND DRAFT ONLY

Safe to schedule

Monday: new papers added to your wiki

Friday: are the course page's links working?

Weekly: a digest of the class's questions

A PERSON MUST ACT

Never schedule

Anything sent to students

Changing marks or attendance

Publishing, deleting, paying

A scheduled agent reads and drafts. A person sends.

BEFORE WE TALK ABOUT USING AI WELL

A routine department task

marks.csv
synthetic marks

→

Attainment agent
does the calculation

→

Report agent
writes the summary

→

HOD summary
what the HOD reads

THE BRIEF I GAVE IT

```
Compute CO attainment for Unit 4 from marks.csv.
A student attains a CO if they score at least 6% on its questions.
A CO is attained if 60% of students attain it.
Write a one-page summary for the HOD.
```

Find the mistake. Compare it with the brief.

TO: HOD · FROM: DEPARTMENT AGENT

Unit 4: course outcome attainment

CO1	96%	**Attained ✓**
CO2	94%	**Attained ✓**
CO3	98%	**Attained ✓**
CO4	95%	**Attained ✓**
CO5	97%	**Attained ✓**

Summary: excellent results; every course outcome attained.

Trace it back: one typo, four careful agents

HOD summary	“every CO attained”: looks perfect
↓	
Report agent	copied the table faithfully
↓	
Attainment agent	computed correctly…
↓	
The brief	…with 6% where I meant 60%: the real mistake

Why it happened

A typo in the brief

A number that looks valid is never questioned

Each agent trusts the one before

The polished format hides it

What would have caught it

Check one CO by hand: a student with 40% marked “attained”

A reviewer rule: flag any threshold below 40%

Gate 2: a person signs, not the agent

USING AI WELL

How much checking does this task need?

Two things decide it: how much the output affects a student, and how hard it is to undo.

POLL · ANSWER BY NUMBER

Where does each task belong?

- 01 Slide drafts
- 02 A practice quiz
- 03 Feedback on a lab report
- 04 Internal marks entry
- 05 A reference list in your paper
- 06 Recommending a student for a scholarship

Quadrants: 1 skim and go · 2 review every item · 3 verify before it goes out · 4 a person decides

The higher the stakes, the more a person decides

	EASY TO UNDO	HARD TO UNDO
HIGH EFFECT ON A STUDENT	**Review every item** check each one before it is sent • Feedback on a lab report	**A person decides** AI may prepare; a person decides and signs • Internal marks entry • Scholarship recommendation
LOW EFFECT	**Skim and go** fix mistakes when you spot them • Slide drafts • Practice quiz	**Verify before it goes out** check every item; mistakes become public • Reference list in your paper

Human in the loop: three levels

Level	What you do	In tonight's studio
In the loop	Approve each step	Gate 1: you approve the plan
On the loop	Watch, review the output	You read review.md
A person decides	AI prepares, you own it	Gate 2: nothing reaches students without you

One design rule: put the gates in before you start.

USING AI WELL

Do and don’t

Do

Give it your own material; ask for citations

Write “done means” before you start

Start in “ask every time” mode

Keep a person at every gate

Check one result by hand

Tell students how you used AI

Don’t

Let it decide marks, admissions or discipline

Paste real student data into a free tool

Trust a citation you haven’t opened

Let an agent send, publish or delete on its own

Use an output you can’t explain

Present AI work as entirely your own

Never put this into an AI tool

- ✕ Student names with marks, roll numbers or attendance
- ✕ Phone numbers, addresses, Aadhaar or any ID number
- ✕ Health, disability or counselling information
- ✕ A manuscript you are reviewing for a journal
- ✕ Passwords, OTPs, API keys
- ✕ Your institution's confidential documents: salaries, HR, legal

DPDP Act, 2023: student data is personal data. Use synthetic or anonymised data, or a tool your institution has approved for it.

Be careful here

Be careful with	The check
Citations and facts that look right	Open the source
Formulas, numbers and units	Work one example by hand
Feedback that rewards fluent English, or a name	Rerun it with a different name and style
Textbook text and images	Summarise and cite; never copy
Unreleased exam papers	Only in a tool your institution approves
Instructions hidden in files and web pages	Treat every file as data, not orders
Students (and you) thinking less	Mark the explanation, not only the answer

The controls you have

Control	What it does	Where you met it
Instruction file	Standing rules, read every time	Day 4: the chip
Skills	How to do a task well	Day 4: the cartridge
Permissions	What needs your yes	Day 4
Hooks	Block risky actions, always	Day 4
Spec / PRD	Decide before anything is built	Day 5 and tonight
Read-only reviewer	A second reader that can’t edit	Tonight
Gates	Nothing moves without you	Tonight
Files and logs	A record of every step	Tonight

Five practices, each with a check

privacy	no real student data in free tools; synthetic or anonymised only
integrity	students and faculty declare AI use; mark the process, not only the output
fairness	rerun the same work with a different name or English style; marks must not change
attribution	every fact traced to a source; nothing copied without credit
a record	keep the plan, review files and logs, so you can explain every decision

A six-line department AI policy

01 **Permitted:** drafting, explanation, practice material, code help, all disclosed

02 **Not permitted:** AI deciding marks, admissions or discipline; AI work passed off as one's own

03 **Data:** no personal student data in tools the institution has not approved

04 **Disclosure:** students attach an AI-use note; faculty follow journal policy

05 **Assessment:** add viva, in-class work and evidence of process

06 **Review:** revisit every semester

DEMO

What the studio made

WHAT THE STUDIO MADE

What we just saw

Plan first	A grilling, then a PRD
Many agents	Three researchers, a debate, four builders, at the same time
You decide	A read-only reviewer and two gates

CLOSE · POSTED TOMORROW MORNING

Your kit

`PRD.md`	the studio, ready to hand to any agent
`agent files`	built tonight, ready to copy
`four skills`	grill-me · pptx · pdf · education skills
`one-page guide`	run the studio on your next class

Poll: which will you try first? The studio · question paper · literature review · CO attainment

CLOSE

Your next 30 days

01 **Week 1:** run the studio on one class you teach

02 **Week 2:** build one more application from the gallery

03 **Week 3:** write one skill for a format your department uses

04 **Week 4:** take the six-line policy to a department meeting

Pick one row for next week. Not all four.

CLOSE · WITHIN ONE WEEK OF TONIGHT

Portfolio and certificate

4a · a teaching tool

4b · a NotebookLM pack

4c · one drafted skill

5a · a plan or spec

5b · your LLM wiki

6a · the studio, on your own class

6b · your policy draft and 30-day plan

The rule

At least five of the seven pieces, posted in the shared space within one week.

06

Thank you

Brief clearly. Check the work. Stay in charge.

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